

Template Week 1 – Bits & Bytes

Student number: 587890

Assignment 1.1: Bits & Bytes intro

What are Bits & Bytes?

A bit is a 0 or a 1

A byte is 8 bit

What is a nibble?

4 bits.

What relationship does a nibble have with a hexadecimal value?

One hexadecimal digit is a nibble

Why is it wise to display binary data as hexadecimal values?

Each digit is 4 bits making it easier to read for humans

3F A2 00 VS 001111111010001000000000

What kind of relationship does a byte have with a hexadecimal value?

one byte is two hex digits.

An IPv4 subnet is 32-bit, show with a calculation why this is the case.

$4 \text{ octets} \times 8 \text{ bits per octet} = 32 \text{ bits}$

Assignment 1.2: Your favourite color

Hexadecimal color code:

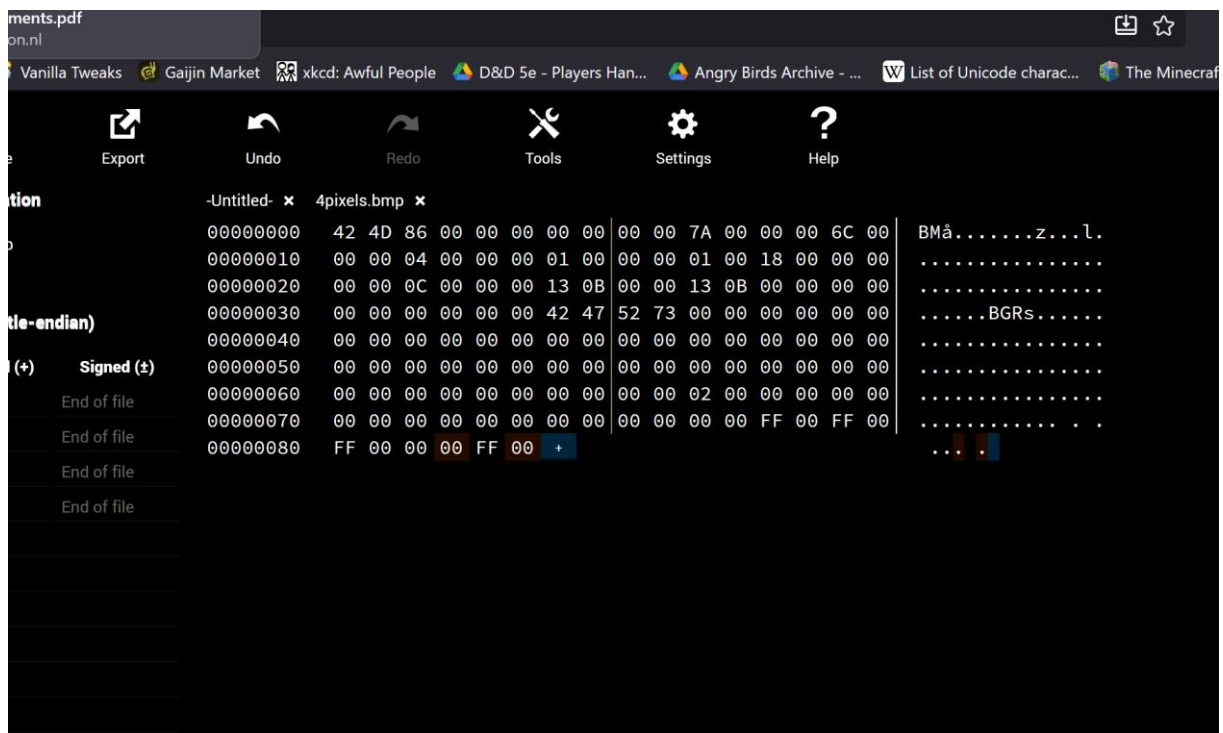
#00ff00

0, 255, 0

Assignment 1.3: Manipulating binary data

Color	Color code hexadecimal (RGB)	Big Endian	Little Endian
RED	255,0,0	#ff0000	0000ff
GREEN	0, 255, 0	#00ff00	00ff00
BLUE	0,0,255	#0000ff	ff0000
WHITE	255,255,255	#ffffff	ffffff
Favourite (previous assignment)	0, 255, 0	#00ff00	00ff00

Screenshot modified BMP file in hex editor:



Assignment 1.4: Student number to HEX and Binary

Convert your student number to a hexadecimal number and a binary number.

Explain in detail that the calculation is correct. Use the PowerPoint slides of week 1.

Division	Remainder
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587,890 ÷ 16	2
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36,743 ÷ 16	7
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2,296 ÷ 16	8
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143 ÷ 16	15
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8 ÷ 16	8
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0x8F872

8 1000

F 1111

8 1000

7 0111

2 0010

10001111100001110010

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